

Noah Mok

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EDUCATION

University of Waterloo | Waterloo, ON

Bachelor of Applied Science, Mechanical Engineering | Sep. 2026 - Apr. 2031

TECHNICAL SKILLS

CAD: Solidworks, Siemens NX, AutoCAD

Coding Languages: Python, C++, VBA, HTML, CSS

Technical Knowledge: Dimensioning and Tolerancing, Solid Modeling, Surface Modeling, Data Analysis, Statics and Kinematics

PROJECTS

Solar Car Canopy | Oct 2026

- Designed a solar car canopy using **Surface Modeling** in **Solidworks** as part of an onboarding challenge for the University of Waterloo Midnight Sun team.
- Crafted an aerodynamic canopy to meet dimensioning constraints, and researched previous aerobody designs to optimize for canopy drag area, reducing drag by more than **50 percent**.

Rotary Selector Switch and Mount | Sep 2026

- Designed a rotary selector switch and its mount using **Siemens NX** as part of an onboarding challenge for the University of Waterloo Automobile Futures Team.
- Examined an engineering drawing to design a part meeting **dimensioning, safety, and manufacturing** constraints for integration into a production vehicle.
- Improved safety, accessibility, and lifetime of the switch while reducing costs by **20 percent**.

Kinematics Calculator | Jul 2026

- Developed a desktop GUI application through **Python** to solve 1D and 2D kinematic equations to help students with complex physics calculations.
- Utilized **SymPy** to dynamically substitute known values and solve reduced equation systems, saving upwards of **15 minutes per calculation**.

Pendulum Simulation | Jul 2026

- Created a real-time animation of pendulum motion using **Matplotlib** to help students properly visualize simple harmonic motion.
- Embedded pendulum motion equations and learned how different parameters can affect motion through **Python**.
- Implemented an efficient solution in less than 50 lines of code in **O(n) time and O(1) memory**.

EXPERIENCE

Applied Mathematics Researcher | Jun 2019, Jun 2022

Epsilon Math Camp & MathPath

- Lead a team of 16 people in researching isoperimetric sets of English words with Prof. Chip Morgan using **data analysis and Python** to reverse engineer the process behind online tools like auto-correct, published in *Parabola*, Vol. 59, Issue 2.
- Co-developed a solution on Watt's linkage motion with Dr. Zoltan Kovacs, published in the *International Journal of Technology in Mathematics Education*, Vol. 27, No. 4.

Student Page | Sep 2023 – Oct 2025

Edmonton Public Library | Edmonton, AB

- Worked 10-12 hours a week organizing, shelving, and sorting book returns and arrivals and assisting customers in navigating software such as Word, Powerpoint, and Outlook, saving individual patrons **up to 30 minutes** from searching books and handling technical services.
- Mentored incoming student pages into daily operations using my experience to maintain the strength of the student page team, and increased efficiency of the student page team **by 50 percent**.

Student Council Vice President | Jan 2025 – Dec 2025

Davidson Academy Online

- Organized school-wide events including a Career Fair, Talent Show, Game Night, and Winter Book Exchange to rejuvenate an online student community, boosting participation **by 15 percent**.
- Led a team of 7 in event planning and outreach, invited guest speakers and club leaders to school events, and represented the student body to the school administration.